# BrickPlan: A Benchmark for Visual Assembly and Error Recovery Supplementary Material

Anonymous CVPR submission

Paper ID Draft

## A. Artifact Scope and Reproducibility

The manuscript describes an evidence snapshot dated 21 October 2026. All earlier models are included through their current versions. Three banks coexist: the 3,292-question white-baseplate static bank, the 1,872-episode dynamic mechanism bank, and the 96-case community module suite. Their counts are not independent model populations, and their experimental results are not interchangeable. The new route study runs on the community module suite, not on the historical six-part car. The complete module suite has no formal learned-agent evaluation or independent human assembly experiment in this snapshot.

The artifact records original model files, source metadata, certificates, cases, reference manuals, intervention witnesses, and order-regression checks. Each instruction contains the physical part and the resulting scene. Image manifests bind the current benchmark renders to their source states. Paper figure provenance separately records source hashes, cameras, native render dimensions, and exported figure hashes. The deterministic study records the four input files for each object and their SHA-256 digests. Re-execution does not require a model API.

The public object gallery is an author-facing inspection tool: it shows fault names and correction witnesses. It is not a valid blind agent input. Evaluation must use the observation adapter, which exposes the permitted images and action handles while retaining certificates, intervention labels, event timing, and recovery witnesses privately. Publishing such annotations supports reproducibility but also makes these public examples unsuitable as an undisclosed held-out test set.

## B. Complete Model Coverage

The three tracks contain 5,260 current records: 3,340 static and 1,920 dynamic. Their 240 configuration entries are not independent objects. All 127 earlier static configurations and all 30 earlier dynamic structures remain in the current banks. The original chair, table, car, and crane are included in Figure S1; another 38 complete object designs and 26 structural configurations are also shown. Twelve community modules extend this coverage rather than replacing it. The artifact's full catalogue lists every model identifier, source, target-part count, case count, and error distribution. Tables at the end of this supplement enumerate all 160 static configurations.

## C. Exact Observation and Action Roles

Table S2 distinguishes the images that define a task from those that show its current state. In a static question, the acceptance target is an intermediate stage. A full-completion reference is provided for context and must not silently replace that stage target. In a dynamic episode, the acceptance target is the completed module and remains fixed even when a hidden intervention occurs.

**Views and handles.** The module interface provides isometric, rear, and top views at $900 \times 660$ pixels. Part and candidate handles are randomized by seed and do not expose target labels. The manual stays accessible throughout the episode. Current-state images do not include the paper's error boxes or corrective arrows. Explicit request and response logs distinguish inspection from manipulation and retain the actual view delivered to the agent.

**One physical action per turn.** An attach action chooses one rack piece and one catalog candidate. A detach action removes one installed piece. The executor checks the resulting configuration and a straight insertion or extraction path. An invalid physical attempt consumes a manipulation attempt, reports a legality code, and leaves the world unchanged. Inspection does not advance construction or trigger an event. Finish requests termination; success additionally requires the exact permitted geometric endpoint. The manipulation budget is $4N + 12$ for $N$ normal module pieces. The outer model runner has a separately configured decision cap so repeated inspection cannot run indefinitely. Both limits must be reported in a model comparison.

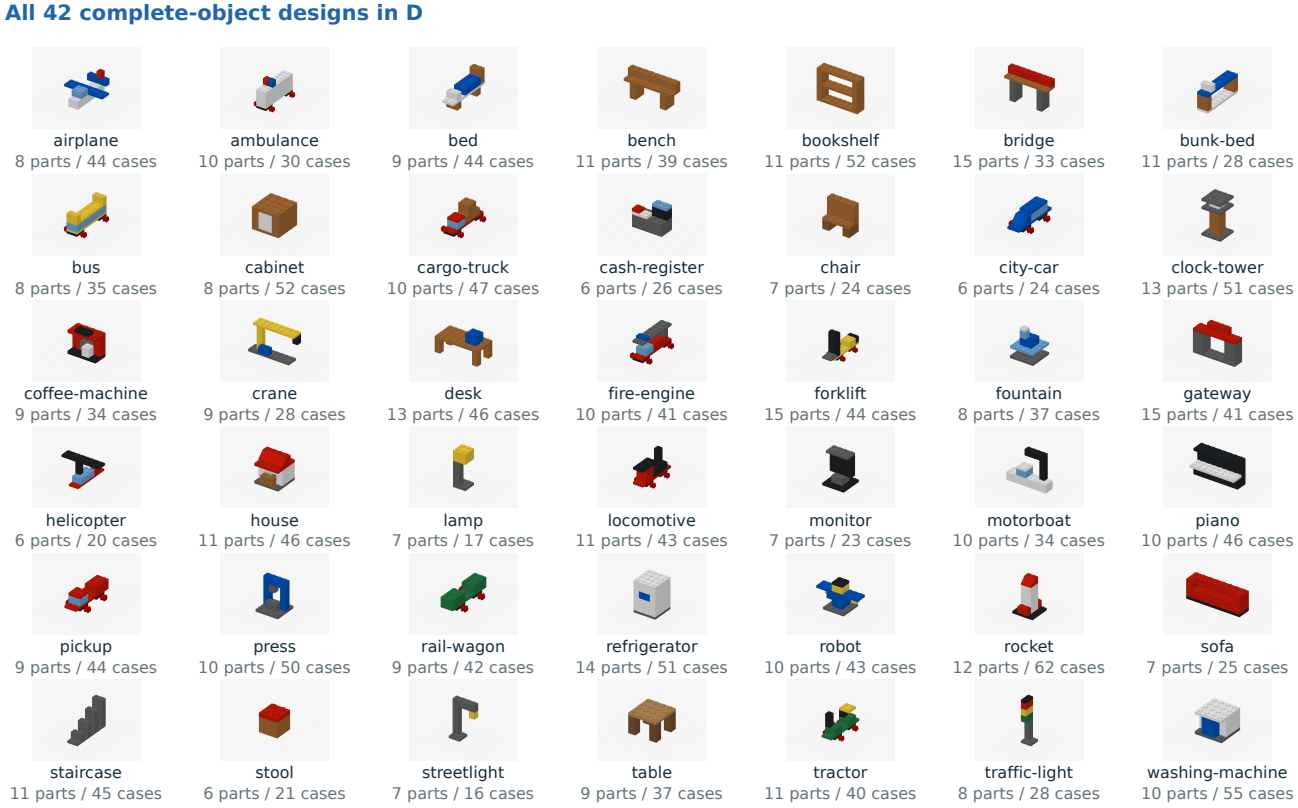


Figure S1. **All 42 complete-object designs in D.** Every object is shown with its target-part count and current episode count. These are deliberately simplified complete constructions, including the original four objects. Their geometric simplicity is not evidence of measured human or model difficulty.

Figure S2. **All 26 structural configurations in D.** These configurations provide source-derived assemblies and controlled tests of access, support, buried errors, and reconstruction. Together with Figure S1, they cover all 68 dynamic structures and 1,872 episodes.

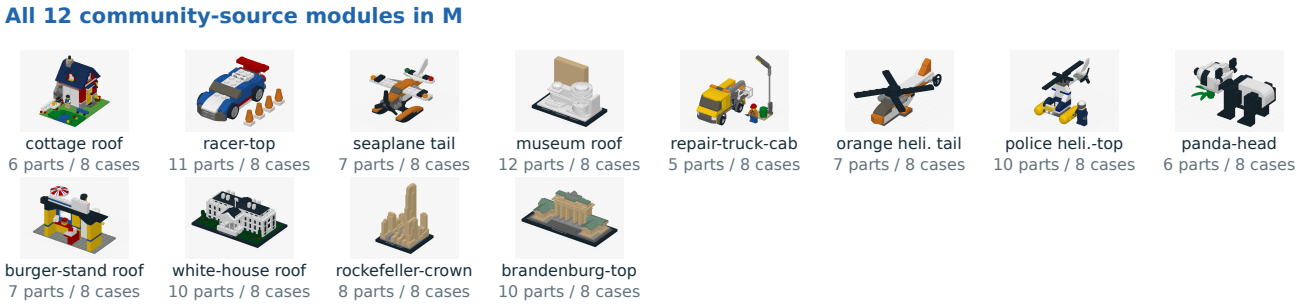


Figure S3. **All twelve community sources in M.** The entire source scene provides context; only the bounded 5–12-part module is manipulated. Part counts below the scenes count those task pieces, not the complete source object. Each module supplies eight cases.

Table S1. **Related evaluation contracts.** Inputs and outputs determine what a failure can mean. This is a task comparison, not a performance comparison.

| Work | Goal information | Output / interaction | Recovery setting |
|---|---|---|---|
| BrickNet; BrickBench | Generation conditions or text | Assembly program / agentic design | Validity during generation; no prescribed external fault in a supplied module. |
| MEPNet; SCANet | Visual assembly manual | Part poses / step correction | Manual execution or correction of a particular assembly step. |
| Break and Make | Interactive inspection of a target | Inspect, disassemble, reconstruct | Reconstruction after a dedicated inspection phase. |
| AsgardBench; RoboRecover | Task and visual scene | Interactive actions / robot policy | Plan adaptation or continuation from execution deviations. |
| BRICKPLAN module suite | Prebuilt context, goal, full remaining manual | One attach, detach, inspect, or finish | Continuous construction with an actual-state intervention; any legal order is accepted. |

**Finite action candidates.** The action catalog is part of the task definition. This is not unrestricted six-degree-of-freedom pose prediction. Candidate previews and legality codes reduce the planning burden and can leak useful physical information, even though neither states the correction. They must be held constant across compared agents; an ablation can quantify their contribution. A symbolic state or expected next-state overlay is privileged information and should be reported as a separate condition.

**Indistinguishable parts.** Terminal comparison uses physical part geometry, color, and pose rather than instance identity. Supported rotational equivalences are limited to plain rectangular parts with the appropriate geometric symmetry. Decorated, directional, and asymmetric pieces are not assumed invariant to arbitrary rotations. Identical-part exchanges are tested separately from sequence exchanges: changing an identifier alone must not create an apparent alternative physical trajectory.

## D. Intervention Selection and Causal Comparisons

**Deterministic adaptive selection.** Each community fault condition names one of three mechanisms and an installed-part threshold. After an accepted manipulation reaches the threshold, the selector inspects the actual state. For a missing part or color replacement, it considers older installed normal pieces, retaining the newest piece to avoid merely undoing the last action. For an extra layer, it searches compatible exposed positions for the spare plate. The control and all three fault conditions share the same inventory and candidate catalog.

An event is admitted only if it changes the state through legal primitives and has a feasible inverse returning to the pre-event state. A replacement uses a same-shape alternative-color spare: the environment removes the original piece and attaches the spare between observations. These two environment primitives do not allow the agent to submit two manipulations in one turn. If no suitable event is available, selection is deferred. At most one event is applied in the current module suite. The private log records the realized location and its inverse.

Table S2. **Image roles in the community interface.** A reference image is not a report of current execution. These roles are explicit in the adapter rather than inferred from file order.

| Role | Content | Interpretation |
|---|---|---|
| Scene overview | Full original object, initial and completed views | Establishes the recognizable object and the supplied context. |
| Module start | Cropped scene before any task manipulation | The initial movable-part state; the context is already assembled. |
| Full-completion reference | Completed remaining module | Explains the whole module; may differ from a static question's acceptance target. |
| Acceptance target | Required final state for this episode | Stage target for static repair; completed module for dynamic completion. |
| Instruction part / after | Isolated part and scene after its reference placement | One executable demonstration; the numbered order is not compulsory. |
| Inventory card | Shape and color of an available piece | Identifies an action handle without declaring its correct placement. |
| Before last action | Actual scene before the agent's last action | Temporal evidence; explicitly absent at the first decision. |
| Current scene | Actual state after execution and any intervention | The authoritative observation of the world now. |
| Candidate preview | Hypothetical placement of a selected piece | A planning aid; not executed, not a correctness certificate. |

**What feasibility establishes.** A legal inverse is a recovery witness, not a proof that it is the shortest recovery or that the error is visually obvious. Every selected normal route in the protocol study has a complete replay, so restoring its pre-event state permits the remaining route to finish. This guarantee does not extend to an arbitrary agent state that was already incorrect before the intervention. Likewise, a reference event's RGB check does not certify visibility at every location an adaptive selector can reach. New trajectories require their own observation-level audit.

**Matched-prefix controls.** For a causal recovery comparison, execute a normal prefix once, save the environment and conversation, and branch it into control and fault continuations. The two branches have identical history before the event. Keep the target, action vocabulary, inventory, renderer, budgets, and model settings fixed. Report both completion and whether the fault actually occurred. A run that finishes correctly without encountering its scheduled fault is a completion, not a successful repair trial.

A conditional recovery rate excludes untriggered episodes from its denominator, so it must be accompanied by exposure coverage and the number of stalled prefixes. Even with the same input seed, two fresh API runs are not necessarily matched-prefix controls: their actions may diverge before the event. Adaptive location selection also creates trajectory-dependent fault difficulty. Stratify by the realized edit and report the reference-location subset separately when appropriate.

## E. Community Objects and Route Diagnostics

Table S3 lists all twelve objects. Original instance counts include the supplied surrounding scene, figures, and accessories. The module count measures the actual task; it does not imply reconstructing the entire source object. The full scenes contain 2,135 source instances; the bounded modules contain 99 normal placements. Larger source models therefore need not be harder tasks.

**Selection of the alternative routes.** Each reference route is the verified manual. An alternative is obtained by testing swaps and retaining a physically legal route with all three feasible adaptive faults. When available, the search favors a route whose geometry at the threshold differs from the reference checkpoint. The study therefore uses selected regression examples, not a random sample of all topological orders or an estimate of agent route choice. Table S4 provides the precise reference-position indices.

**Exact-prefix opportunity.** The comparison inspects the otherwise normal physical trajectory for equality with the reference checkpoint. This avoids confounding trigger eligibility with a different repair policy. It yields 36/36 eligible reference-route fault trials and 9/36 alternative-route trials. The actual-state selector applies an event in all 36 trials in each group. A separate comparator requiring every intermediate state to match the manual accepts 12/12 reference routes and 0/12 alternatives. The terminal comparator accepts all 24 normal routes. These numbers establish counterexamples to an overly restrictive contract, not a learned-

Table S3. **Complete community suite.** Every row has four static and four dynamic cases: control, missing piece, wrong color, and extra layer. The threshold is a count of installed normal pieces, not a compulsory reference prefix.

| Source set | Object | Remaining module | Source parts | Task parts | Threshold |
|---|---|---|---|---|---|
| 31009 | Cottage | Roof section | 270 | 6 | 3 |
| 31027 | Blue racer | Upper cabin | 67 | 11 | 3 |
| 31028 | Seaplane | Tail assembly | 53 | 7 | 3 |
| 21004 | Guggenheim Museum | Roof section | 208 | 12 | 4 |
| 3179 | Repair truck | Cabin top and lights | 128 | 5 | 4 |
| 30181 | Orange helicopter | Tail assembly | 52 | 7 | 3 |
| 30311 | Police helicopter | Rotor assembly | 59 | 10 | 4 |
| 30026 | Panda | Head section | 61 | 6 | 4 |
| 6683 | Burger stand | Roof and sign | 79 | 7 | 3 |
| 21006 | White House | Chimneys and roof details | 556 | 10 | 4 |
| 21007 | Rockefeller Center | Upper crown | 240 | 8 | 7 |
| 21011 | Brandenburg Gate | Top crest | 362 | 10 | 6 |
| Total | 12 objects | 96 cases | 2,135 | 99 | — |

Table S4. **Alternative legal routes used in the deterministic study.** Reference-position numbers identify pieces; the actual executor uses randomized handles. All routes reach the same normal target. The threshold states differ for nine objects.

| Object | Alternative route in reference-position indices | Different threshold state |
|---|---|---|
| Cottage | 1, 2, 4, 3, 5, 6 | Yes |
| Racer | 1, 6, 3, 4, 5, 2, 7, 8, 9, 10, 11 | Yes |
| Seaplane | 1, 6, 3, 4, 5, 2, 7 | Yes |
| Museum | 1, 2, 3, 5, 4, 6, 7, 8, 9, 10, 11, 12 | Yes |
| Repair truck | 1, 2, 3, 5, 4 | Yes |
| Orange helicopter | 1, 3, 2, 4, 5, 6, 7 | No |
| Police helicopter | 1, 2, 3, 5, 4, 6, 7, 8, 9, 10 | Yes |
| Panda | 1, 2, 3, 5, 4, 6 | Yes |
| Burger stand | 2, 1, 3, 4, 5, 6, 7 | No |
| White House | 1, 2, 3, 5, 4, 6, 7, 8, 9, 10 | Yes |
| Rockefeller Center | 1, 2, 3, 5, 4, 6, 7, 8 | No |
| Brandenburg Gate | 1, 2, 3, 4, 5, 7, 6, 8, 9, 10 | Yes |

agent performance improvement.

The blind policy follows its normal route after any change. It completes 24/24 controls and 0/72 fault trials. The privileged policy uses the recorded inverse immediately after the event and then continues; it completes 96/96 trials. No model API is involved. A privileged repair trace tests solvability and logging, not visual detection, reasoning, or human-level assembly skill.

## F. Static Migration and Error Coverage

The static bank uses an explicit white baseplate and disallows unsupported floating states or passage through the board. Its migration ledger records the five routes in Table S5. All resulting repair witnesses replay under the new executor. Figure S4 shows actual current and target states for eight error categories.

Of the 267 replacements, 250 source targets were not buildable under the new rules, 13 had board-penetrating initial states, and four were duplicate states. Replacements preserve their source links and substitution reasons. The ledger's 3,025 preserved cases include ten repaired initial scenes; "preserved" therefore does not mean that every initial coordinate was unchanged. The 160 controls have a zero-action repair witness.

Table S5. **White-baseplate migration ledger.** Counts refer to questions, not independent objects.

| Migration method | Questions |
|---|---|
| Direct replay | 2,144 |
| Verified dismantle/rebuild | 834 |
| Witness using held components | 37 |
| Corrected board-penetrating initial scene | 10 |
| Replaced incompatible source case | 267 |
| Total certified questions | 3,292 |

Table S6. **Dynamic mechanism counts.** Each episode belongs to one recorded family, including the explicitly compound family.

| Condition | Episodes |
|---|---|
| Normal control | 160 |
| Removed piece | 214 |
| Shifted piece | 246 |
| Cascaded removal | 160 |
| Extra layer | 179 |
| Early cover | 31 |
| Wrong color | 218 |
| Wrong size | 80 |
| Rotated piece | 132 |
| Swapped positions | 23 |
| Wrong part type | 98 |
| Buried color error | 209 |
| Compound fault | 122 |
| Total | 1,872 |

A complete geometric replay is a necessary check, but does not establish that every question has sufficient visual evidence or comprehensible wording. The full static RGB package and per-question human readability review are still pending. It would be incorrect to interpret the 3,292/3,292 replay count as a human-validated visual dataset size.

## G. Dynamic Mechanisms and Visibility

The mechanism bank contains 68 structures: 42 simplified recognizable objects and 26 smaller structural configurations. Of its 1,872 episodes, 1,588 use complete-object models and 284 use the remaining configurations. There are 160 controls and 1,712 interventions across twelve families (Table S6). This is broader mechanism coverage than the three error families in the community suite, but does not substitute for testing those mechanisms on the larger source-preserving modules.

**Machine visibility screens.** Pixel differences use a maximum RGB-channel change of at least 24. The reference mechanism audit retains 1,553 interventions as primary machine candidates with at least 2,000 contrast pixels in an available view. Another 136 interventions are classified as action cancellation, and 23 have low contrast; there are no duplicate event states. A visible change in pixel space is not proof that a person can identify the incorrect part or infer a valid correction. Re-rendering and rechecking the realized event is necessary when a different route changes its location or occlusion.

The community reference snapshots use a separate minimum of 128 changed pixels in at least one fixed module view. All 36 reference fault snapshots pass that machine screen. Its lower threshold is an engineering admission criterion, not an experimentally calibrated human perceptual threshold. The two visual admission rules must not be pooled as one validated population.

**Splits and difficulty.** The mechanism bank groups sources into 43 families for development and evaluation separation. Its balanced development subset has 108 faults (nine per family) and 42 controls; the evaluation subset has 168 faults (fourteen per family) and 72 controls, where "family" in these counts means an error mechanism. Source-family separation applies within this bank, not across every artifact in the repository. The community suite is a public development collection and has no undisclosed held-out source-object evaluation in the current report.

Part count, prerequisite depth, access obstruction, and the amount of reconstruction needed after a fault describe different dimensions of difficulty. Neither source-file size nor the labels "roof" and "car" justify an easy/medium/hard ranking without performance evidence. Figure S6 reports case coverage without making that inference.

## H. Historical Development Evidence

**Earlier API car pilot.** A prior interface version records three completed episodes with the provider route named `google/gemini-4-argon` on a six-part car (Table S7). These were development traces with seed 41 and a 40-decision cap. They used a different bank and interface, without the community manuals, and cannot populate a result table for the current suite. One control response was invalidly formatted; no illegal physical action was executed. A normal-run trajectory diverged from the fault runs before the intervention, so these traces are not a paired causal estimate of repair difficulty.

**How to read the GT/model comparison.** The main paper's eight-column filmstrip uses the extra-layer and control traces from this archive. GT is a feasible reference, not a required intermediate trajectory. The extra-layer run completes in fourteen decisions with seven accepted physical moves; its first two attachments and final two additions have a different order from GT. In the control run, P001 is legally placed at C006 rather than its target pose, then detached and reattached correctly. Physical legality therefore does not imply target correctness. All displayed episodes eventually succeed; the red cell marks a transient wrong placement, not final failure.

The figure export checks every saved state hash and feedback code in both runs (32 decisions total), and supplies a response-free JSON with the original certificate, selected states, source hashes, and actions. The plotted states use one shared camera and framing. Inspections are omitted; the final control image also follows an unshown last place-

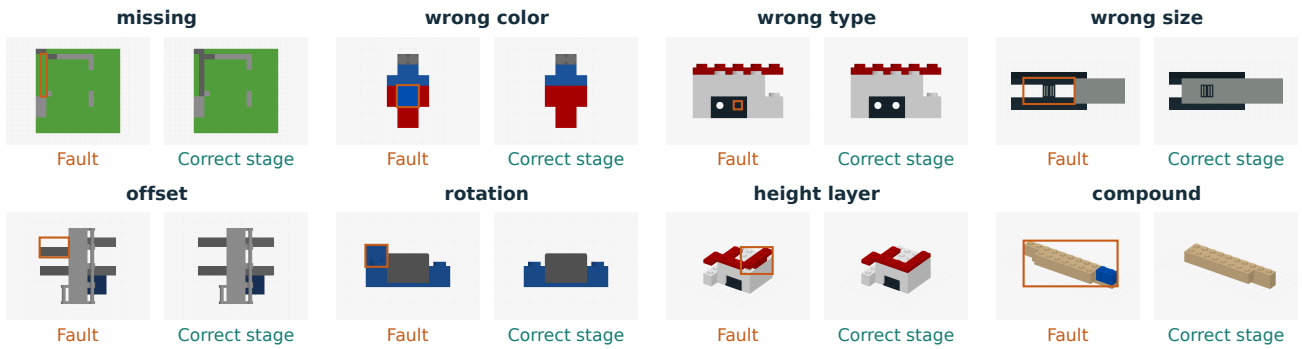


Figure S4. **Eight static error categories on the white baseplate.** Each compact cell places the faulty current state on the left and the correct stage target on the right. Categories cover missing pieces, color/type/size substitutions, translation, rotation, layer height, and compound errors. Orange difference boxes are paper annotations only.

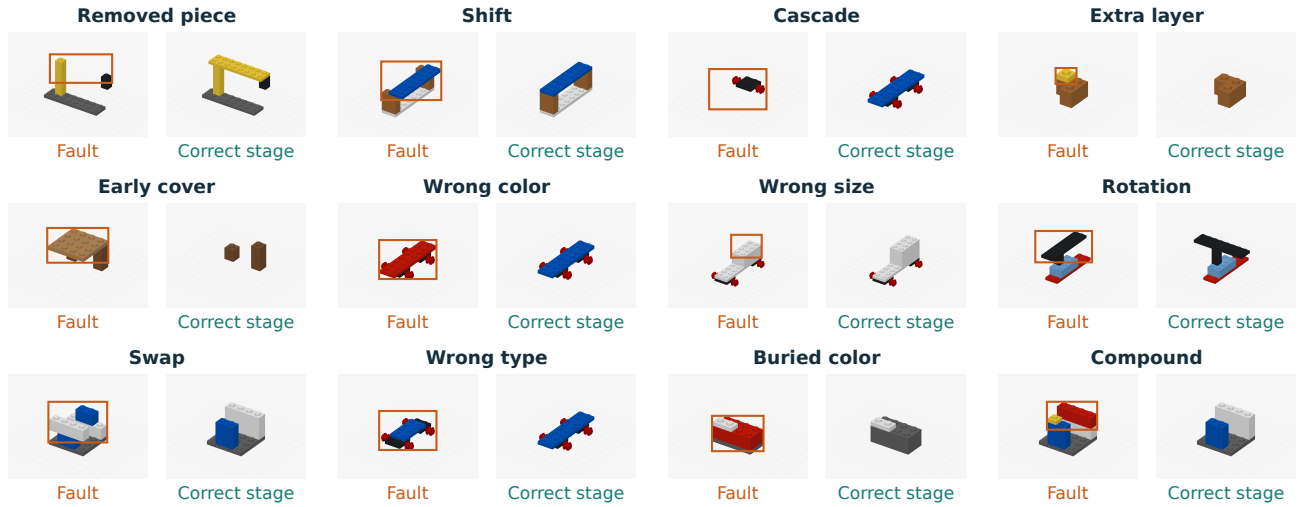


Figure S5. **All twelve dynamic error mechanisms.** Each cell compares a faulty state with its correct intermediate stage, using a common view. Examples are drawn from complete objects in D. They illustrate detectable scene changes and witnessed repairs; they do not establish human error frequencies or model detection performance. Boxes are explanatory annotations only.

Table S7. **Archived pilot only.** Costs are the recorded charges for these individual calls, not current prices or a projected experiment budget. All three completed the older car task.

| Condition | Decisions | Legal moves | USD |
|---|---|---|---|
| Control | 18 | 8 | 0.1950043 |
| Extra layer | 14 | 7 | 0.2022685 |
| Removed piece | 14 | 7 | 0.2140114 |

ment. The external extra layer is labeled separately from the model's mistake. These examples illustrate reporting conventions and cannot establish current-bank success rates or an independent effect of interventions.

Earlier no-progress runs and HTTP 403 provider failures occurred during interface development. An API access failure is not evidence of a model's assembly ability. There are zero paid model episodes in the new community suite and mechanism-v6 release record.

**Earlier human check.** One recorded participant answered binary change-detection questions correctly on eleven selected, unprompted images; a twelfth item had already exposed its answer and is excluded. Two further reviewers were reported to agree, but have no independent submission records. This is not a three-rater study, a timed assembly baseline, or validation of all current cases. It supports only a narrow development observation about those eleven images.

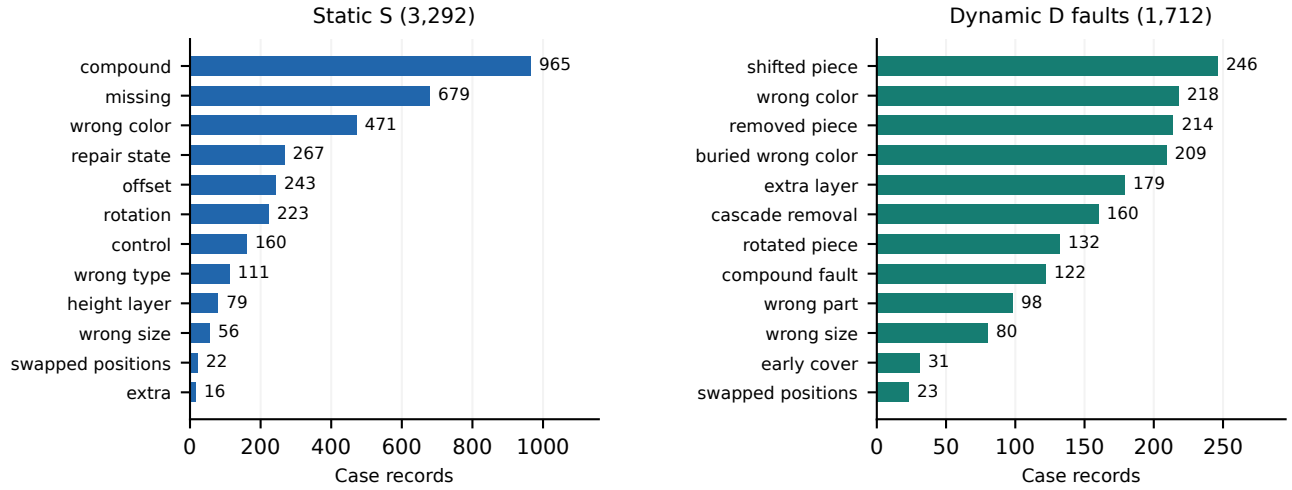


Figure S6. **Coverage of the broader banks.** Static counts include normal controls; dynamic bars show intervention families and exclude 160 controls. Counts describe repeated tasks on shared models, not independent object diversity or measured difficulty.

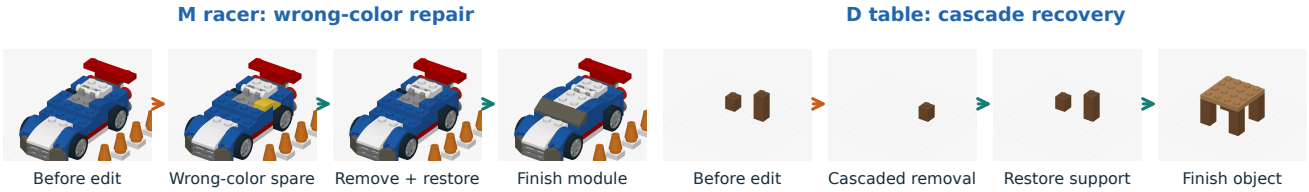


Figure S7. **Two certified repair examples, eight views in one row.** Left four: replace a wrong-color racer cabin piece and finish the community module. Right four: restore pieces removed by a cascade in the original table object. Each transition summarizes selected stages; any multi-part recovery executes separate one-brick actions. These are geometry witnesses, not model rollouts.

# I. Planned Experiments and Reserved Result Tables

All dashes in Tables S8–S12 are uncollected measurements. They are neither zero scores nor executor results. The artifact contains machine-readable case assignments, participant schedules, and a results template whose empty values are `null`. These define the planned experiment; they are not an implementation claim for the paired API runner or human study.

**Models and primary samples.** Use three commercial visual-model families (V1–V3) and two accessible open-weight sizes (V4–V5). Record exact checkpoint/provider identifiers, revision/date, image budget, prompt, and decoding settings before execution. Apply seeds 41, 42, and 43 to handle randomization and repeated runs; a provider seed is recorded only when supported. The S test list contains 467 question IDs from 20 source groups. Migration changed the source of some questions; 151 stale assignments are repaired while retaining the existing source-group allocation. The corrected current split contains 2,286 train, 539 development, and 467 test questions. D uses its existing 168-fault/72-control evaluation definitions, with fourteen faults per mechanism. M uses all 48 static cases, 36 dynamic fault definitions, and twelve dynamic controls; it remains a public development panel. The full-bank secondary report covers every model configuration and retains separate track/source averages. No joint tuning may use a source later presented as unseen in another track.

**Prerequisites and budgets.** Complete image rendering and blind readability review before S visual scoring. Verify the observation adapter, trajectory saving, actual-event logging, and both cloned continuations on deterministic smoke tests before paid trials. For this planned comparison, use a decision cap $\max(40, 6N + 20)$ for every track, with $N$ target pieces; retain M's additional $4N + 12$ manipulation cap. Repeated inspection and malformed outputs consume decisions. S's feasibility witnesses remain reference costs, not optimality claims. Store input images, actual actions, accepted/rejected moves, termination reason, and service errors. Retry transient transport failures at most twice with

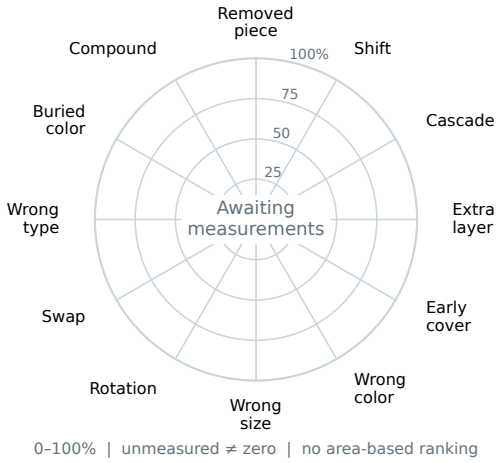


Figure S8. **Error-specific recovery profile template.** Each axis is conditional recovery for one D mechanism in Table S8. All axes use the same 0–100% scale. Missing measurements have no curve; an untriggered stratum is undefined, not zero. Report exposure counts and source-clustered intervals in the table.

the same request; do not retry semantically incorrect model answers into success.

**Pair construction and exposure.** At the first admissible fault opportunity, save the environment and visible conversation once. Continue a control copy without an edit and a fault copy with the prescribed actual-state edit. A separate control run with the same seed is not this pair. D's 72 control definitions are additional construction controls; each of its 168 fault jobs still needs its own cloned control continuation. Keep scheduled faults with no admissible prefix in the exposure denominator, and report their stopping reasons. Count every API call in both continuations. The base plan has 7,725 static attempts and 4,320 dynamic shared-prefix or normal jobs across five models and three seeds (12,045 work units, not API calls). There are 3,060 fault jobs, each requiring two continuations when the prefix is reached. Ablations, the human-comparison panel, visual admission, and full-bank coverage are additional work.

**Reading the radar profiles.** The aggregate profile in the main paper separates tracks and outcomes; Figure S8 separates error families within D. Plot one series per model only when all required estimates exist, without filling missing values or joining across missing axes. Both templates are bound to the same result fields as their tables and currently contain no model curves. Rates use source-macro aggregation; their denominators differ across tasks, so neither polygon area nor an average of axes is a score. Exposure is an opportunity for repair, not evidence that the error was recognized. Tables remain the source of counts and uncertainty.

**Human comparison without written responses.** Select twelve S control/fault pairs from different sources, twelve D pairs covering the twelve mechanisms, and one pair per M module, cycling evenly through its three errors. This yields 36 pairs or 72 conditions. Six independent participants each operate one condition from every pair, using buttons and graphical actions. Each condition receives three observations; nobody sees both members of the same pair. Randomize order and balance normal/fault assignments across participants. The exact schedule is saved before collection. Train on separate practice scenes. Record completion, exposure, actions, time, and an optional "cannot see / cannot operate" button. This is not three independent records when only one person submits an answer. The same panel is evaluated with V1–V5 under the same views, action limits, and seeds. Human estimates additionally cluster by participant.

**Ablations and analysis units.** Instantiate each main-paper ablation independently for V1 and V5, chosen by model slot before viewing outcomes. Apply no-manual only to M, no-previous-view only to dynamic tracks, and keep all other inputs and cases fixed. The privileged symbolic condition may expose actual part poses but must not silently provide intervention labels or recovery actions. Recompute both control and fault continuations when input changes alter the prefix. Report source-macro completion and recovery, as well as aggregate exposure numerator/denominator; do not average undefined stratum rates as zero. Use 2,000 bootstrap replicates clustered by source, retaining seeds and case variants together; paired differences resample the same sources. Sparse categories remain descriptive.

Do not infer a detection miss solely from an incomplete object. If anomaly judgments are measured, run a separately declared judgment panel at recorded observations without interrupting the primary assembly runs, and retain no-change controls. A state-query ablation and action trace inspection distinguish visible evidence, planning, and action selection, but their causal scope depends on what information changes. Estimate money from a small current-interface pilot for each selected model. Old car charges are archived observations, not current price quotes. The experiment guide specifies caps and a cost formula; no paid calls are made in this revision.

## J. Rendering, Attribution, and Physical Limits

**Publication images.** All brick scenes in the paper are rendered from original geometry at $3200 \times 2400$ pixels, with $2\times$ supersampling in each dimension before downsampling.

Table S8. **Reserved D mechanism results.** Report exposed/assigned counts and exposure-conditioned recovery with source-clustered 95% intervals. The primary design has fourteen fault definitions per row and three seeds per model. Human measurements belong to the separate common panel, not these denominators.

| Mechanism | Definitions | V1 R | V2 R | V3 R | V4 R | V5 R |
|---|---|---|---|---|---|---|
| Removed piece | 14 | — | — | — | — | — |
| Shifted piece | 14 | — | — | — | — | — |
| Cascaded removal | 14 | — | — | — | — | — |
| Extra layer | 14 | — | — | — | — | — |
| Early cover | 14 | — | — | — | — | — |
| Wrong color | 14 | — | — | — | — | — |
| Wrong size | 14 | — | — | — | — | — |
| Rotation | 14 | — | — | — | — | — |
| Swapped positions | 14 | — | — | — | — | — |
| Wrong part type | 14 | — | — | — | — | — |
| Buried color error | 14 | — | — | — | — | — |
| Compound fault | 14 | — | — | — | — | — |

Table S9. **Reserved human/model common-panel results.** All agents use the same 72 conditions. Human cells aggregate three independent observations per condition; model cells use three runs. S: static completion; C/T/R: dynamic normal completion, intervention coverage, and recovery; FA: unnecessary modification of a normal control; time is median active seconds. Report D and M separately.

| | S | D panel | | | M panel | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Agent | Success | C | T | R | C | T | R | S FA | Time |
| Independent humans | — | — | — | — | — | — | — | — | — |
| V1 | — | — | — | — | — | — | — | — | — |
| V2 | — | — | — | — | — | — | — | — | — |
| V3 | — | — | — | — | — | — | — | — | — |
| V4 | — | — | — | — | — | — | — | — | — |
| V5 | — | — | — | — | — | — | — | — | — |

Table S10. **Reserved difficulty breakdown.** Instantiate by track and model. Bands describe planned stratification, not a validated easy/hard ranking. Stage is installed target-piece count divided by target size at the realized event. Temporary removals use the recorded recovery witness.

| Stratum | Assigned | Exposed | Recovery |
|---|---|---|---|
| 4–7 target parts | — | — | — |
| 8–10 target parts | — | — | — |
| 11–15 target parts | — | — | — |
| Early stage ($q \leq 1/3$) | — | — | — |
| Middle ($1/3 < q \leq 2/3$) | — | — | — |
| Late stage ($q > 2/3$) | — | — | — |
| No temporary removal | — | — | — |
| Temporary correct-part removal | — | — | — |

Table S11. **Reserved failure ledger.** Counts are measured separately by track and model. Physical rejection and format errors are attempt-level counters and may co-occur within an episode; they are not an additive partition of all failures.

| Observed event | Count | Denominator |
|---|---|---|
| Provider / transport failure | — | requests |
| Invalid action format | — | decisions |
| Rejected physical action | — | move attempts |
| Scheduled fault never exposed | — | fault jobs |
| Wrong finish / exhausted budget | — | episodes |
| Control completed, fault failed | — | exposed pairs |
| Explicit wrong anomaly judgment | — | valid judgments |

Composite figure exports use 600 dpi. PDF and SVG retain vector labels, arrows, tables, and process boxes. The paper's larger renders do not replace the $900 \times 660$ images received by the current community interface. Orange boxes and manual highlights are presentation annotations and are disclosed in the relevant captions. They are not supplied as automatic fault detections.

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Table S12. **Reserved efficiency and paired-outcome results.** Instantiate separately for S, D, and M; paired recovery is not applicable to S. Paired R is recovery among exposed pairs whose normal branch succeeds, with its denominator shown. Tokens and charges include shared prefixes, both continuations, and billed retries.

| Model | Paired R | Denom. | Decisions | Inspections | Legal moves | Rejected | Seconds | USD |
|---|---|---|---|---|---|---|---|---|
| V1 | — | — | — | — | — | — | — | — |
| V2 | — | — | — | — | — | — | — | — |
| V3 | — | — | — | — | — | — | — | — |
| V4 | — | — | — | — | — | — | — | — |
| V5 | — | — | — | — | — | — | — | — |

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**Physical scope.** The environment checks rigid geometry, supported connections, and straight manipulation paths. Unsupported surrounding parts are retained as conservative collision bounds rather than deleted, which can reject some motions that a more precise mesh would allow. The preassembled context is fixed. The executor does not model deformation, insertion force, frictional stability, accidental slipping, articulated mechanics, or a robot gripper. Consequently, "legal" and "recoverable" refer to this geometric contract. Neither the protocol study nor its renders establish real-robot performance.

Table S13. **Complete static configuration inventory (1/2).** P: target parts; Q: current questions. Identifiers refer to the accompanying full model catalogue.

| Index | Model/configuration | P | Q | Index | Model/configuration | P | Q |
|---|---|---|---|---|---|---|---|
| S001 | 10001 | 11 | 72 | S041 | v23-10233 | 11 | 23 |
| S002 | 10001-deep | 16 | 48 | S042 | v23-10241 | 11 | 18 |
| S003 | 10014 | 11 | 33 | S043 | v23-10248 | 11 | 25 |
| S004 | 10036 | 11 | 73 | S044 | v23-10258 | 11 | 18 |
| S005 | 10036-alt1 | 11 | 28 | S045 | v23-10261 | 11 | 17 |
| S006 | 10128 | 11 | 24 | S046 | v23-10262 | 11 | 22 |
| S007 | 10156 | 11 | 68 | S047 | v23-10267 | 11 | 18 |
| S008 | 10156-alt1 | 11 | 20 | S048 | v23-10271 | 11 | 21 |
| S009 | 10159 | 11 | 23 | S049 | v23-1472 | 11 | 18 |
| S010 | 10213 | 11 | 62 | S050 | v23-1484 | 11 | 16 |
| S011 | 10214 | 11 | 21 | S051 | v23-1489 | 11 | 23 |
| S012 | 10220 | 11 | 28 | S052 | v23-1490 | 11 | 16 |
| S013 | 31009 | 11 | 18 | S053 | v23-1490-alt1 | 11 | 27 |
| S014 | 31027 | 11 | 22 | S054 | v23-1572 | 11 | 22 |
| S015 | 31027-alt1 | 11 | 26 | S055 | v23-1572-alt1 | 11 | 23 |
| S016 | 31027-kart | 11 | 22 | S056 | v23-1632 | 11 | 19 |
| S017 | 31028 | 11 | 6 | S057 | v23-1633 | 11 | 23 |
| S018 | 31028-alt1 | 11 | 25 | S058 | v23-1682 | 11 | 16 |
| S019 | 31028-sailboat | 11 | 25 | S059 | v23-1775 | 11 | 16 |
| S020 | 5867 | 11 | 22 | S060 | v23-1775-alt1 | 10 | 21 |
| S021 | omr-10242 | 11 | 16 | S061 | v23-1792 | 11 | 22 |
| S022 | omr-10265 | 11 | 9 | S062 | v23-1808 | 11 | 22 |
| S023 | omr-10269 | 11 | 3 | S063 | v23-1808-alt1 | 11 | 21 |
| S024 | omr-21309 | 11 | 19 | S064 | v23-1854 | 11 | 18 |
| S025 | omr-31031 | 11 | 20 | S065 | v23-1854-alt1 | 11 | 25 |
| S026 | omr-31088 | 11 | 6 | S066 | v23-20015 | 11 | 21 |
| S027 | omr-42004 | 11 | 18 | S067 | v23-20015-alt1 | 11 | 25 |
| S028 | omr-42020 | 11 | 18 | S068 | v23-21004 | 11 | 14 |
| S029 | omr-42020-alt1 | 11 | 17 | S069 | v23-21005 | 11 | 17 |
| S030 | omr-42054 | 11 | 18 | S070 | v23-21006 | 11 | 22 |
| S031 | omr-42061 | 11 | 18 | S071 | v23-21007 | 11 | 18 |
| S032 | omr-42066 | 11 | 7 | S072 | v23-21009 | 11 | 14 |
| S033 | omr-42102 | 11 | 17 | S073 | v23-21010 | 11 | 22 |
| S034 | omr-42102-alt1 | 11 | 9 | S074 | v23-21011 | 11 | 21 |
| S035 | v23-10037 | 11 | 15 | S075 | v23-21012 | 11 | 22 |
| S036 | v23-10124 | 11 | 22 | S076 | v23-21014 | 11 | 22 |
| S037 | v23-10158 | 11 | 29 | S077 | v23-21016 | 11 | 18 |
| S038 | v23-10184 | 11 | 17 | S078 | v23-21017 | 11 | 22 |
| S039 | v23-10194 | 11 | 17 | S079 | v23-21018 | 11 | 22 |
| S040 | v23-10226 | 11 | 24 | S080 | v23-21020 | 11 | 17 |

Table S14. **Complete static configuration inventory (2/2).** P: target parts; Q: current questions. Identifiers refer to the accompanying full model catalogue.

| Index | Model/configuration | P | Q | Index | Model/configuration | P | Q |
|---|---|---|---|---|---|---|---|
| S081 | v23-21021 | 11 | 22 | S121 | v23-31040 | 11 | 5 |
| S082 | v23-21022 | 11 | 22 | S122 | v23-31040-alt1 | 11 | 26 |
| S083 | v23-21024 | 11 | 22 | S123 | v23-31045 | 11 | 17 |
| S084 | v23-21029 | 11 | 22 | S124 | v23-31048 | 11 | 22 |
| S085 | v23-21030 | 11 | 22 | S125 | v23-31049 | 11 | 7 |
| S086 | v23-21037 | 11 | 22 | S126 | v23-31063 | 11 | 23 |
| S087 | v23-21045 | 11 | 22 | S127 | v23-31066 | 11 | 23 |
| S088 | v23-21302 | 11 | 17 | S128 | v23-31070 | 11 | 8 |
| S089 | v23-21303 | 11 | 21 | S129 | v23-31072 | 11 | 22 |
| S090 | v23-21318 | 11 | 22 | S130 | v23-31072-alt1 | 11 | 22 |
| S091 | v23-21319 | 11 | 23 | S131 | v23-31084 | 11 | 22 |
| S092 | v23-21321 | 11 | 17 | S132 | v23-31092 | 11 | 11 |
| S093 | v23-21325 | 11 | 21 | S133 | v23-31092-alt1 | 11 | 22 |
| S094 | v23-30022 | 7 | 1 | S134 | v23-3178 | 11 | 18 |
| S095 | v23-30023 | 7 | 20 | S135 | v23-3178-alt1 | 11 | 22 |
| S096 | v23-30025 | 11 | 11 | S136 | v23-3179 | 11 | 5 |
| S097 | v23-30025-alt1 | 11 | 20 | S137 | v23-3179-alt1 | 11 | 26 |
| S098 | v23-30026 | 11 | 21 | S138 | v23-3181 | 11 | 18 |
| S099 | v23-30026-alt1 | 11 | 20 | S139 | v23-3221 | 11 | 23 |
| S100 | v23-30027 | 11 | 8 | S140 | v23-3223 | 11 | 22 |
| S101 | v23-30027-alt1 | 11 | 6 | S141 | v23-3223-alt1 | 8 | 18 |
| S102 | v23-30029 | 11 | 22 | S142 | v23-3223-deep | 16 | 27 |
| S103 | v23-30029-alt1 | 11 | 22 | S143 | v23-4018 | 9 | 21 |
| S104 | v23-30181 | 11 | 22 | S144 | v23-6678 | 11 | 22 |
| S105 | v23-30181-alt1 | 11 | 7 | S145 | v23-6678-deep | 16 | 23 |
| S106 | v23-30285 | 11 | 3 | S146 | v23-6679 | 11 | 16 |
| S107 | v23-30311 | 11 | 22 | S147 | v23-6682 | 11 | 22 |
| S108 | v23-30311-alt1 | 11 | 24 | S148 | v23-6682-alt1 | 11 | 21 |
| S109 | v23-30312 | 11 | 21 | S149 | v23-6683 | 11 | 22 |
| S110 | v23-30472 | 11 | 15 | S150 | v23-6683-alt1 | 8 | 18 |
| S111 | v23-30472-alt1 | 11 | 24 | S151 | v23-6689 | 11 | 23 |
| S112 | v23-30542 | 11 | 12 | S152 | v23-6689-alt1 | 11 | 25 |
| S113 | v23-30542-alt1 | 11 | 20 | S153 | v23-6690 | 11 | 22 |
| S114 | v23-30571 | 11 | 22 | S154 | v23-6690-alt1 | 11 | 8 |
| S115 | v23-30571-alt1 | 11 | 25 | S155 | v23-6698 | 11 | 10 |
| S116 | v23-30571-deep | 16 | 16 | S156 | v23-6698-alt1 | 11 | 24 |
| S117 | v23-31025 | 11 | 17 | S157 | v23-6699 | 11 | 17 |
| S118 | v23-31037 | 11 | 18 | S158 | v23-6699-alt1 | 11 | 18 |
| S119 | v23-31038 | 11 | 17 | S159 | v23-6745 | 11 | 22 |
| S120 | v23-31039 | 11 | 23 | S160 | v23-6753 | 11 | 23 |